Week 5 Feasibility Memo

CariSurg MedTech Pathways — AI-Assisted Triage Data Exploration

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Prepared for: Dr. De Fretias and the Mercer General Hospital Emergency Department Board

Project: Caribbean ED Validation of Machine-Learning Triage Support

Dataset: yaleemmlc_admissionprediction_triage.csv

One-Sentence Verdict

This dataset is suitable for building a Week 6 baseline triage model, provided post-triage leakage columns are excluded, rare ESI classes are handled carefully, and results are treated as exploratory until validated in a Caribbean emergency department setting.

Dataset Summary

The dataset contains 55,121 emergency department encounters and 226 columns. These include patient demographics, arrival details, triage vital signs, disposition-related variables, and chief complaint flags. The main target variable is esi, which represents the Emergency Severity Index triage level, where ESI 1 is most urgent and ESI 5 is least urgent.

The dataset is clinically useful because it includes information normally relevant during triage, such as age, gender, race, ethnicity, heart rate, blood pressure, respiratory rate, oxygen saturation, oxygen device use, temperature, glucose, and presenting complaint indicators. The ESI distribution is mainly mid-acuity, with most encounters recorded as ESI 2 or ESI 3. ESI 1 cases are very rare, so the dataset may support baseline modelling but requires careful evaluation of the sickest patients.

Common chief complaints include abdominal pain, chest pain, shortness of breath, back pain, falls, cough, dizziness, vomiting, weakness, altered mental status, alcohol intoxication, and suicidal presentation. These complaints are useful because they include both common emergency department presentations and high-risk warning signs.

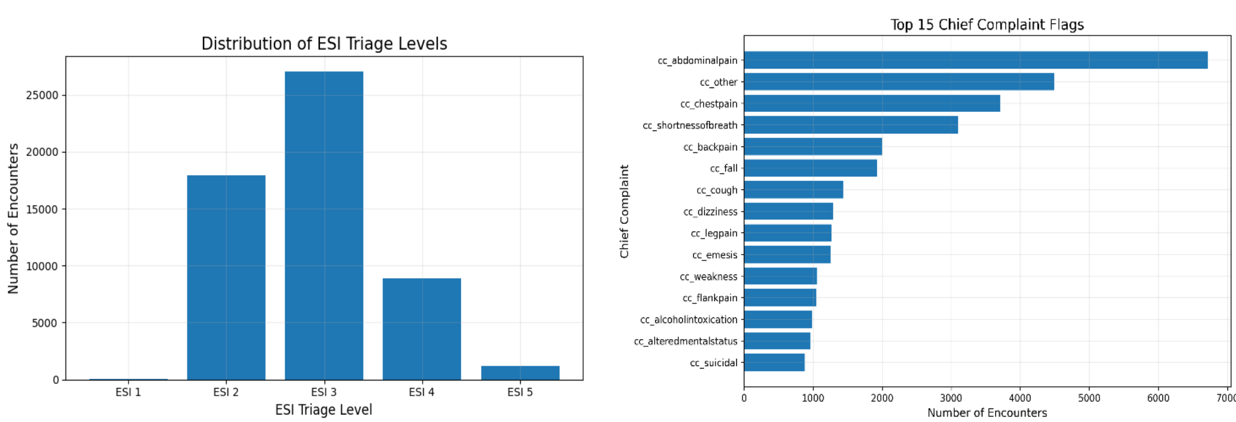

Figure 1:Key Week 5 exploratory findings showing ESI distribution and common chief complaints.

Top 3 Quality Concerns

1. **Class imbalance:** ESI 1 patients are extremely rare compared with ESI 2, ESI 3, and ESI 4 patients. A model could therefore appear accurate overall while still performing poorly on the most urgent cases. In Week 6, performance should be checked separately for high-acuity patients, especially ESI 1 and ESI 2.

2. **Outcome leakage:** Some columns may contain information that would not be available at the time of triage. In particular, disposition and previousdispo may reflect what happened after assessment or after the ED visit. These should be excluded from Week 6 modelling so the model only uses information available during triage.

3. **External validity:** Although the dataset is large and clinically rich, it may not fully represent a Caribbean emergency department. Patient demographics, disease patterns, staffing, workflow, resources, and documentation habits may differ from Mercer General Hospital. Local validation would therefore be required before clinical use.

Top 3 Reasons to Proceed

1. **Real triage target:** The dataset includes the esi triage level, giving Week 6 modelling a clear supervised learning target based on actual triage acuity.

2. **Clinically relevant features:** The dataset includes important triage vital signs such as heart rate, systolic and diastolic blood pressure, respiratory rate, oxygen saturation, oxygen device use, temperature, and glucose. These reflect circulation, breathing, oxygenation, infection risk, and metabolic status.

3. **Rich chief complaint information:** The chief complaint flags describe why patients attended the emergency department. Complaints such as chest pain, shortness of breath, altered mental status, suicidal presentation, weakness, and abdominal pain may help identify clinically important presentations that vital signs alone may miss.

Caveats

This Week 5 analysis does not build or validate a model. It only assesses whether the dataset is suitable for baseline modelling. Visualizations, correlations, and summaries show early data-quality patterns, but they do not prove prediction or clinical usefulness. Correlation should be interpreted cautiously. A feature associated with ESI may not cause the triage level or remain useful in real-world deployment. Some relationships may reflect documentation habits, site-specific workflow, or hidden clinical decisions. Chief complaint flags should also be used carefully because many are uncommon. Very rare complaints may need to be grouped, filtered, or tested carefully during Week 6 modelling. The main caveat is Caribbean external validity. This dataset can support exploration and baseline model development, but it cannot prove that an AI-supported triage model would be safe, fair, or practical at Mercer General Hospital without local validation.

Top-10 Feature Shortlist for Week 6

Table 1: Top 10 feature shortlist for week 6.

Rank	Feature	Reason
1	age	Older patients may carry higher triage risk.
2	triage_vital_o2_device	Oxygen device use suggests respiratory support.
3	triage_vital_o2	Oxygen saturation reflects oxygenation status.
4	cc_chestpain	Chest pain may indicate urgent cardiopulmonary disease.
5	cc_shortnessofbreath	Shortness of breath is a high-risk ED complaint.
6	cc_alteredmentalstatus	Altered mental status often requires urgent assessment.
7	triage_glucose	Abnormal glucose may indicate metabolic instability.
8	triage_vital_hr	Heart rate reflects circulatory stress or illness.
9	triage_vital_rr	Respiratory rate is an early warning sign.
10	triage_vital_sbp	Systolic blood pressure helps flag shock or severe hypertension.

Leakage columns such as disposition and previousdispo should be excluded from Week 6 modelling. This shortlist should be treated as a starting hypothesis, not as proven feature importance.

Final Recommendation

The dataset is strong enough to proceed to a Week 6 baseline triage model. It has a real ESI target, clinically meaningful vital signs, and useful chief complaint information. However, modelling should proceed cautiously, with leakage columns removed, rare ESI classes evaluated separately, and all findings framed as exploratory until tested using Caribbean emergency department data.

Notes and AI Use

This memo is based on Week 5 exploratory analysis of yaleemmlc_admissionprediction_triage.csv and the figures generated in the accompanying Jupyter notebook. No raw patient rows were pasted into AI tools. AI assistance was used to refine structure and wording, while final interpretation and recommendations were reviewed by the student.